

$\langle \cdot \rangle$
 Zobrazení $\alpha: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ $\langle u, v \rangle = u^T A v$
 $a_{11} > 0$ a $\det(A) \neq 0$

$\langle u, u \rangle > 0 \quad u = \begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\langle u, u \rangle = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = a_{11}x^2 + 2a_{12}xy + a_{22}y^2$

zvolíme vektor $u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T$ dosadíme do vzorce $a_{11} \cdot 1^2 + 2a_{12} \cdot 1 \cdot 0 + 0^2 a_{22} = a_{11}$

a proto $a_{11} > 0$

$\det(A) > 0$

kde pro $y=1$ $f(x) = a_{11}x^2 + 2a_{12}x + a_{22}$

$\Delta = (2a_{12})^2 - 4a_{11}a_{22} < 0$

$a_{12}^2 - a_{11}a_{22} < 0 \quad | : -1$

$a_{11}a_{22} - a_{12}^2 > 0$